

BATCH No:MAD31

**FOODGPT: COVERATIONAL AI FOR PERSONALIZED
RESTAURANT AND DISH RECOMMENDATIONS USING
ADVANCED MACHINE LEARNING**

*Major project report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Computer Science & Engineering**

By

SADINENI PAVANI	(21UECS0542)	(VTU20221)
ABBIGANI SESA SAI KALYAN	(21UECM0003)	(VTU20209)
AVULA LAKSHMI BHAGIRADHI	(21UECS0055)	(VTU20230)

*Under the guidance of
Dr. K. RAJATHI, M.Tech., Ph.D.,
ASSOCIATE PROFESSOR*



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

May, 2025

BATCH No:MAD31

**FOODGPT: COVERATIONAL AI FOR PERSONALIZED
RESTAURANT AND DISH RECOMMENDATIONS USING
ADVANCED MACHINE LEARNING**

*Major project report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Computer Science & Engineering**

By

SADINENI PAVANI	(21UECS0542)	(VTU20221)
ABBIGANI SESA SAI KALYAN	(21UECM0003)	(VTU20209)
AVULA LAKSHMI BHAGIRADHI	(21UECS0055)	(VTU20230)

*Under the guidance of
Dr. K. RAJATHI, M.Tech., Ph.D.,
ASSOCIATE PROFESSOR*



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

May, 2025

CERTIFICATE

It is certified that the work contained in the project report titled "FOODGPT: CONVERSATIONAL AI FOR PERSONALIZED RESTAURANT AND DISH RECOMMENDATIONS USING ADVANCED MACHINE LEARNING " by "SADINENI PAVANI (21UECS0542), ABBIGANI SETHA SAI KALYAN (21UECM0003), AVULA LAKSHMI BHAGIRADHI (21UECS0055)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Signature of Supervisor

Dr. K. Rajathi, M.Tech., Ph.D.,

Associate Professor

Computer Science & Engineering

School of Computing

Vel Tech Rangarajan Dr. Sagunthala R&D

Institute of Science and Technology

Signature of Head/Assistant Head of the Department

Dr. N. Vijayaraj/Dr. M. S. Murali dhar

Professor & Head/ Professor & Assistant Head

Computer Science & Engineering

School of Computing

Vel Tech Rangarajan Dr. Sagunthala R&D

Institute of Science and Technology

Signature of the Dean

Dr. S P. Chokkalingam

Professor & Dean

School of Computing

Vel Tech Rangarajan Dr. Sagunthala R&D

Institute of Science and Technology

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

SADINENI PAVANI

Date: / /

ABBIGANI SESA SAI KALYAN

Date: / /

AVULA LAKSHMI BHAGIRADHI

Date: / /

APPROVAL SHEET

This project report entitled FOODGPT: CONVERSATIONAL AI FOR PERSONALIZED RESTAURANT AND DISH RECOMMENDATIONS USING ADVANCED MACHINE LEARNING by SADINENI PAVANI (21UECS0542), ABBIGANI SETHA SAI KALYAN (21UECM0003), AVULA LAKSHMI BHAGIRADHI (21UECS0055) is approved for the degree of B.Tech in Computer Science & Engineering.

Examiners

Supervisor

Dr.K. Rajathi, M.Tech., Ph.D.,
Associate Professor,.

Date: / /

Place:

ACKNOWLEDGEMENT

We express our deepest gratitude to our **Honorable Founder Chancellor and President Col. Prof. Dr. R. RANGARAJAN B.E. (Electrical), B.E. (Mechanical), M.S (Automobile), D.Sc., and Foundress President Dr. R. SAGUNTHALA RANGARAJAN M.B.B.S., Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, for her blessings.**

We express our sincere thanks to our respected Chairperson and Managing Trustee **Mrs. RANGARAJAN MAHALAKSHMI KISHORE, B.E., Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, for her blessings.**

We are very much grateful to our beloved **Vice Chancellor Prof. Dr. RAJAT GUPTA**, for providing us with an environment to complete our project successfully.

We record indebtedness to our **Professor & Dean , School of Computing, Dr. S P. CHOKKALINGAM, M.Tech., Ph.D., & Professor & Associate Dean , School of Computing, Dr. V. DHILIP KUMAR, M.E., Ph.D.,** for immense care and encouragement towards us throughout the course of this project.

We are thankful to our **Professor & Head, Department of Computer Science & Engineering, Dr. N. VIJAYARAJ, M.E., Ph.D., and Associate Professor & Assistant Head, Department of Computer Science & Engineering, Dr. M. S. MURALI DHAR, M.E., Ph.D.,** for providing immense support in all our endeavors.

We also take this opportunity to express a deep sense of gratitude to our **Dr. K. RAJATHI, M.Tech., Ph.D.,** for her cordial support, valuable information and guidance, she helped us in completing this project through various stages.

A special thanks to our **Project Coordinators Dr. SADISH SENDIL MURUGARAJ, Professor, Dr. S. RAJAPRAKASH, M.E., Ph.D., Mr. V. ASHOK KUMAR, B.E., M.Tech.,** for their valuable guidance and support throughout the course of the project.

We thank our department faculty, supporting staff and friends for their help and guidance to complete this project.

SADINENI PAVANI	(21UECS0542)
ABBIGANI SESA SAI KALYAN	(21UECM0003)
AVULA LAKSHMI BHAGIRADHI	(21UECS0055)

ABSTRACT

FOODGPT is a conversational AI system developed to deliver highly personalized restaurant and dish recommendations using advanced machine learning and natural language processing techniques. Unlike conventional recommendation platforms that rely on static filters and limited personalization, FOODGPT engages users through natural dialogue to understand specific preferences such as cuisine type, dietary restrictions, budget, and mood. The system is structured into three main modules like Data Collection and Preprocessing, which gathers real-time data from APIs like Zomato and cleans it for analysis. Model Loading and Personalization, where hybrid recommendation techniques combining collaborative and content-based filtering alongside transformer based NLP models generate context-aware suggestions. Backend Integration with a Streamlit Web App, which provides a user-friendly interface for seamless interaction. FOODGPT leverages sentiment analysis and continuous user feedback to refine recommendations over time, adapting to evolving tastes and behavioral patterns. The model achieves a 90 percent accuracy rate using Random Forest algorithms, significantly outperforming traditional decision tree based systems. It also ensures low latency with response times under two seconds, even for complex queries. By offering dish-level insights and real time updates on restaurant data, FOODGPT enhances the user's dining experience, transforming how people discover food and making it a valuable tool for both consumers and the food service industry.

Keywords: Restaurant Discovery, Dish Recommendation, Hybrid Recommendation Engine, Content-based Filtering, Collaborative Filtering, Real-time Data Integration, User Preferences, Sentiment Analysis, AI Chatbot, Zomato API, Dynamic Menu Insights

LIST OF FIGURES

4.1	System Architecture For FoodGPT	14
4.2	Data Flow Diagram For FoodGPT	15
4.3	Use Case Diagram For FoodGPT	16
4.4	Class Diagram For FoodGPT	17
4.5	Sequence Diagram For FoodGPT	18
4.6	Activity Diagram For FoodGPT	19
5.1	Login Form For FoodGPT	24
5.2	Registration Form For FoodGPT	25
5.3	Test Image	27
7.1	vtu20230 offer letter	32
7.2	vtu20221 offer letter	32
7.3	vtu20209 offer letter	33
10.1	Plagiarism Report	39

LIST OF TABLES

6.1 **Analysis Table** 30

LIST OF ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CV	Computer Vision
GPT	Generative Pre-Trained Transformer
ML	Machine Learning
NLP	Natural Language Processing
RF	Random Forest
UI	User Interface

TABLE OF CONTENTS

	Page.No
ABSTRACT	v
LIST OF FIGURES	vi
LIST OF TABLES	vii
LIST OF ACRONYMS AND ABBREVIATIONS	viii
1 INTRODUCTION	1
1.1 Introduction	1
1.2 Background	2
1.3 Objective	2
1.4 Problem Statement	3
2 LITERATURE REVIEW	4
2.1 Existing System	6
2.2 Related Work	7
2.3 Research Gap	7
3 PROJECT DESCRIPTION	8
3.1 Existing System	8
3.2 Proposed System	9
3.3 Feasibility Study	10
3.3.1 Economic Feasibility	10
3.3.2 Technical Feasibility	10
3.3.3 Social Feasibility	11
3.4 System Specification	11
3.4.1 Tools and Technologies Used	12
3.4.2 Standards and Policies	12
4 SYSTEM DESIGN AND METHODOLOGY	14
4.1 System Architecture	14

4.2	Design Phase	15
4.2.1	Data Flow Diagram	15
4.2.2	Use Case Diagram	16
4.2.3	Class Diagram	17
4.2.4	Sequence Diagram	18
4.2.5	Activity Diagram	19
4.3	Algorithm & Pseudo Code	19
4.3.1	Algorithm	19
4.3.2	Pseudo Code	20
4.4	Module Description	21
4.4.1	Data Collection and Preprocessing EDA Details	21
4.4.2	Model Loading and Personalization Engine Details	22
4.4.3	Backend Integration and Streamlit Web App Details	22
4.5	Steps to implement the project	22
4.5.1	Dataset Collection and Preprocessing:	22
4.5.2	Model Training and Fine-Tuning:	23
4.5.3	Real-Time Prediction and Web App Integration:	23
5	IMPLEMENTATION AND TESTING	24
5.1	Input and Output	24
5.1.1	Input Design	24
5.1.2	Output Design	25
5.2	Testing	25
5.3	Testing Strategies	26
5.3.1	Unit Testing	26
5.3.2	Integration Testing	26
5.3.3	System Testing	26
5.3.4	Performance Evaluation	27
6	RESULTS AND DISCUSSIONS	28
6.1	Efficiency of the Proposed System	28
6.2	Efficiency of the Proposed System	29
6.3	Comparison of Existing and Proposed System	29
6.4	Comparative Analysis-Table	30
6.5	Comparative Analysis-Graphical Representation and Discussion	30

7	INDUSTRY DETAILS	31
7.1	Industry name:	31
7.1.1	Duration of Internship:	31
7.1.2	Duration of Internship in months:	31
7.1.3	Industry Address:	31
7.2	Internship offer letter	32
7.3	Internship Completion certificate	33
8	CONCLUSION AND FUTURE ENHANCEMENTS	34
8.1	Summary	34
8.2	Limitations	34
8.3	Future Enhancements	35
9	SUSTAINABLE DEVELOPMENT GOALS (SDGs)	36
9.1	Alignment with SDGs	36
9.2	Relevance of the Project to Specific SDG	37
9.3	Potential Social and Environmental Impact	37
9.4	Economic Feasibility(Costs)	38
10	PLAGIARISM REPORT	39
11	SOURCE CODE	40
11.1	Source Code	40
	References	42

Chapter 1

INTRODUCTION

1.1 Introduction

In the age of digital convenience, the way individuals discover and choose restaurants or dishes has significantly transformed. With a simple tap on a smartphone, users can explore thousands of dining options, view ratings, read reviews, and access menus. However, as the food industry continues to grow rapidly and user expectations evolve, conventional restaurant recommendation platforms struggle to keep up with the demand for personalization, contextual relevance, and real-time adaptability. Most existing systems rely heavily on static filters such as cuisine, location, or star ratings while ignoring deeper user-specific preferences, past dining behavior, dietary restrictions, mood, or even social contexts. This gap often leads to user dissatisfaction, repetitive suggestions, and missed culinary experiences. To address these challenges, FOODGPT proposes an intelligent and conversational approach to food discovery. Built on the foundations of Conversational AI, NLP, and ML, FOODGPT offers a next-generation solution that transforms the way users interact with food recommendation platforms.

Instead of navigating through layers of menus and filters, users can simply talk to FOODGPT, just like chatting with a knowledgeable foodie friend. They can express complex queries like “Find me a cozy place with the best gluten-free chocolate cake under 10 Dollars nearby,” and FOODGPT will respond with context aware, tailored suggestions. Using NLP, the system can interpret user intent in natural language, capturing subtle nuances such as cravings, moods, and dietary needs. Through ML algorithms, the platform refines its recommendations over time, delivering hyper-personalized suggestions that adapt as user tastes evolve. In summary, FOODGPT revolutionizes the restaurant and dish discovery process by offering a conversational, intelligent, and real-time solution. It moves beyond basic filtering mechanisms, creating a deeply engaging and intuitive user experience. With its advanced AI capabilities, FOODGPT not only saves users time but also enhances their culinary journeys by matching them with food experiences that truly resonate with their preferences,

lifestyle, and budget. As dining becomes increasingly personal and digital, platforms like FOODGPT represent the future of food discovery and decision-making.

1.2 Background

In today's digital age, discovering new restaurants and dishes has become a crucial part of the modern dining experience. While numerous restaurant recommendation platforms exist, they often provide generic results based on basic filters such as location, cuisine, or user ratings. These traditional systems lack personalization, fail to consider evolving user preferences, and often do not provide real-time updates about restaurants.

Additionally, users frequently struggle to find specific dish recommendations, as most platforms focus solely on restaurant suggestions without offering insights into popular or well-reviewed menu items. To bridge this gap, FOODGPT leverages Conversational AI, NLP, and ML to provide highly personalized restaurant and dish recommendations. By integrating real-time restaurant data and user preferences, FOODGPT aims to enhance the dining experience, ensuring that users receive tailored recommendations that align with their unique tastes, dietary restrictions, and budget constraints.

1.3 Objective

The rapid growth of AI-driven recommendation systems in industries such as e-commerce, entertainment, and travel has demonstrated the power of personalization in enhancing user engagement and satisfaction. However, restaurant recommendation platforms have largely remained static, relying on pre-defined filters and generic ranking mechanisms. Another key motivation is the inconvenience of manual restaurant search processes. Most existing platforms require users to navigate through a long list of restaurants, reviews, and menus, making it difficult to make quick and informed decisions. Users often have to read multiple reviews, cross-check menus, and consider different sources before finalizing their choice. Moreover, they rarely receive specific dish recommendations an aspect that plays a significant role in enhancing the dining experience.

FOODGPT simplifies this by using NLP-based chatbot interactions, allowing users to express their preferences naturally while receiving highly targeted restau-

rant and dish suggestions, reducing decision fatigue and improving the overall user experience. Restaurant closures, menu changes, price fluctuations, and shifts in customer sentiment can render traditional recommendation platforms outdated and unreliable. Many users end up visiting restaurants with incorrect information due to a lack of real-time updates. By integrating real-time restaurant review APIs, FOODGPT ensures that all recommendations remain up-to-date, preventing users from experiencing misleading or inaccurate suggestions. This real-time adaptability makes FOODGPT a powerful, future-proof solution for both casual diners and food enthusiasts looking for reliable and tailored restaurant recommendations.

1.4 Problem Statement

In the current digital landscape, restaurant discovery and food recommendation systems are often limited by static search filters, generalized suggestions, and a lack of personalization. These systems typically focus on recommending restaurants based on basic parameters such as location, cuisine, or user ratings, without fully understanding or adapting to the individual preferences, dietary needs, or contextual circumstances of users. Additionally, most platforms fail to offer dish-specific recommendations, leaving users uncertain about what to order even after selecting a restaurant.

Moreover, the absence of real-time data integration means that users may receive outdated or inaccurate information about restaurant hours, menu availability, pricing, or dish popularity. The inability to interact with these platforms in a natural, conversational manner further complicates the user experience, especially for those who are unfamiliar with certain cuisines or have specific food-related requirements. These limitations result in an impersonal and inefficient discovery process, leading to suboptimal dining decisions and user dissatisfaction. As users increasingly seek tailored experiences in every aspect of their lives, there is a pressing need for an intelligent system that can understand and respond to natural language queries, adapt recommendations dynamically, and provide actionable, dish-level insights in real time.

Chapter 2

LITERATURE REVIEW

[1] Y. Chen et al.,(2021) proposed the work models for food recommendation as a constrained question-answering problem over a food knowledge graph. By combining structured dietary knowledge with user preferences, the model produces highly relevant recommendations. It is designed to handle large-scale datasets and multi-faceted constraints efficiently. This study is a key reference for developing scalable, intelligent food recommendation systems. Its integration of semantic reasoning is directly applicable to FOODGPT's goal of conversational intelligence.

[2] S. Park et al.,(2022) proposed a recipe recommendation system tailored to users' dietary preferences and restrictions. It uses a hybrid recommendation engine that balances user taste profiles and nutritional concerns. The model improves satisfaction by filtering out ingredients users dislike or cannot consume. This approach provides a solid foundation for building personalized, health-conscious food advisors. It complements FOODGPT's aim of tailoring suggestions through deep preference modeling.

[3] X. Liu et al.,(2023) developed a neural network architecture that combines collaborative filtering and content-based features for recipe recommendations. The model learns latent user-food interactions and adapts recommendations based on evolving preferences. It achieves high accuracy and personalization by leveraging neural embeddings. The method is beneficial for FOODGPT's recommendation module where dynamic preference learning is crucial. It also addresses the cold-start problem common in new-user scenarios.

[4] H. P. Haritha et al.,(2023) introduced a rule-based chatbot that suggests food based on user preferences and basic health input. While not built on deep learning, it highlights how conversational interfaces can simplify user interaction. The system performs basic food filtering using ingredient rules and user intents. It shows the feasibility of dialogue-driven food advising for small-scale implementations.

This serves as an early prototype example for FOODGPT’s conversational flow.

[5] S. Agarwal et al.,(2024) proposed an AI-based recommender system that captures user preferences from prior food choices and feedback. The model uses sentiment analysis and history tracking to improve accuracy. It adapts dynamically with each interaction, enhancing personalization. The system’s design supports user engagement through intelligent, evolving recommendations. It informs FOODGPT’s use of feedback loops for interactive learning and real-time suggestions.

[6] Z. Zhang et al.,(2024) proposed MOPI-HFRS which is a personalized, health-aware food recommendation framework using multi-objective optimization and large language models (LLMs). It balances between taste, nutritional value, and health restrictions. The system interprets complex user inputs and explains its suggestions, adding transparency. Its use of LLMs enables contextual reasoning similar to human conversation. This makes it an ideal reference for building FOODGPT’s intelligent and interpretable recommendation system.

[7] X. Ju et al.,(2022) introduced MenuAI which is a transformer-based model that predicts restaurant dish preferences by analyzing contextual embeddings. The system performs well even in sparse or cold-start conditions, which are common in food platforms. It captures dish-level semantics and user behavior more effectively than traditional models. The model’s use of deep contextual learning aligns closely with FOODGPT’s goal of advanced personalization. It also shows the power of transformers in food-related NLP tasks.

[8] A. Smith et al.,(2023) explored a context-aware food recommender system that includes time, location, and user mood. Using deep learning, it adapts suggestions based on real-world variables, enhancing recommendation relevance. The model dynamically changes its output based on user context and trends. It supports the idea of situationally aware food suggestions for mobile or voice assistants. This is crucial for FOODGPT’s adaptability in varied dining contexts.

[9] M. Johnson et al.,(2024) introduced this paper incorporates nutritional metrics like calories, macronutrients, and dietary goals into a personalized recommendation engine. It uses user health data to generate food suggestions that align with medical

advice. The system provides an intelligent diet planning framework that can serve both regular users and patients. It enhances FOODGPT by showing how AI can personalize food choices based on health needs. Nutritional integration is vital for building responsible food AI systems.

[10] L. Wang et al.,(2022) introduced this paper and reviews the evolution of conversational AI in restaurant recommendation, comparing traditional rule-based bots to deep learning models. It identifies gaps in personalization, dialogue quality, and context understanding. The review covers chatbot architectures, NLP techniques, and emerging technologies in the domain. It is foundational for understanding where FOODGPT fits in the research landscape. The insights help guide the design of a more natural and effective food conversation agent.

2.1 Existing System

Existing restaurant recommendation systems, such as Zomato, Yelp and Google Maps, primarily rely on static filters and general algorithms to provide suggestions based on factors like location, cuisine, price range, and average user ratings. While these platforms offer convenience and basic discovery features, they lack the depth of personalization and adaptability needed to cater to individual user preferences. Most of these systems do not account for real-time variables such as current restaurant occupancy, menu availability, or dynamic pricing. Furthermore, they seldom offer detailed, dish-specific recommendations, leaving users to depend on vague reviews or guesswork when selecting meals.

These platforms also fail to support natural language queries, making the user experience rigid and less intuitive. Users must navigate through multiple layers of drop-downs, filters, and menus to find relevant results, which can be time-consuming and inefficient. In addition, current systems provide limited support for dietary restrictions or nuanced user requirements such as preferred ambiance, special ingredients, or mood-based dining suggestions. As a result, the current landscape of restaurant and food discovery remains largely generic and transactional, lacking the intelligent, personalized, and conversational interface that modern users increasingly demand.

2.2 Related Work

Over the past few years, several studies and systems have been developed to enhance the quality of restaurant and food recommendations through artificial intelligence, natural language processing, and machine learning. These approaches have significantly advanced the domain by addressing challenges in recommendation accuracy, user personalization, and real-time responsiveness.

Chakraborty et al. 2023 proposed an AI-powered food recommendation engine that used content-based filtering along with collaborative filtering to recommend dishes based on user preferences and previous ratings. Their system incorporated user profile data, including taste preferences and dietary restrictions, to enhance personalization. The study highlighted the need for more adaptive models that could evolve with user behavior over time, making recommendations more relevant with continued use.

Mandal and Sharma 2022 introduced a chatbot-based restaurant recommendation system that utilized natural language processing for query understanding. The chatbot provided recommendations based on location and cuisine type. However, it was limited in its ability to recommend specific dishes or process complex multi-intent queries, which are critical for comprehensive dining experiences.

2.3 Research Gap

Despite extensive research on AI-driven food recommendations, several gaps remain unaddressed:

Limited Conversational AI for Personalized Food Recommendations

- Most existing systems rely on predefined filters and do not engage users in natural conversations to refine recommendations dynamically.

Lack of Dish-Level Recommendations

- While restaurant suggestions are common, very few platforms recommend specific dishes based on user preferences, reviews, and popularity.

Absence of Real-Time Data Integration

- Many existing platforms fail to incorporate real-time restaurant data, leading to outdated and inaccurate recommendations.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

In the current landscape, several restaurant recommendation systems and food delivery applications dominate the market, such as Zomato, Swiggy, Yelp, Uber Eats, and Google Maps. These platforms assist users in discovering restaurants based on location, ratings, cuisines, and reviews. While these systems offer basic filtering options such as type of cuisine, cost per person, delivery time, and distance, they lack advanced personalization and contextual understanding. Most of these platforms follow a static, one-size-fits-all approach to recommendations, which often leads to generic suggestions that do not align with individual user preferences, dietary needs, or changing taste profiles.

The existing systems heavily rely on traditional filtering techniques and user ratings. For example, a user might search for "Italian restaurants near me" and the system would return a list sorted by popularity, rating, or proximity. Although helpful, this method fails to consider deeper personalization aspects such as a user's specific taste (e.g., spicy or mild food), allergy constraints (e.g., gluten-free, nut-free), or mood-based preferences (e.g., romantic ambiance, quick lunch spots). As a result, users often miss out on discovering hidden culinary gems within a restaurant's menu. In summary, while existing restaurant and food discovery platforms provide a foundation for basic dining recommendations, they fall short in terms of personalization, real-time awareness, dish-level insights, and interactive user experiences.

Disadvantages:

- **Lack of Personalization:** Most platforms fail to consider deeper, personalized factors such as user preferences, dietary restrictions, mood, or social context. As a result, the recommendations may not align with users' unique tastes, often leading to dissatisfaction.
- **Dish-Specific Recommendations:** Existing systems focus primarily on restaurant-level recommendations and rarely suggest specific dishes based on

individual preferences. This misses an essential part of the dining experience, as many users know what they want to eat but struggle to find the best place to get it.

- **Limited Real-Time Data:** Traditional systems are not integrated with real-time restaurant data such as availability, menu changes, wait times, or seasonal promotions. This results in outdated recommendations and missed opportunities for users.

3.2 Proposed System

The proposed system, FOODGPT, aims to revolutionize restaurant and dish recommendations by combining the power of Conversational AI, Natural Language Processing, and Machine Learning. Unlike existing platforms that offer static and generic suggestions, FOODGPT is designed to deliver a dynamic, personalized, and intelligent user experience through a chatbot interface. The system architecture is organized into six well-structured phases, ensuring a logical and efficient workflow throughout the development process. In the first phase, Research and Requirement Analysis, a comprehensive study of user expectations, existing systems, and literature will be conducted to define the scope and feature set of the chatbot. This is followed by the Data Collection and Preprocessing phase, where restaurant data including menus, reviews, and user ratings will be collected using APIs like Zomato. This data will be preprocessed to ensure it is clean and ready for use in NLP models.

During the Model Development phase, advanced machine learning techniques will be applied to build both content-based and collaborative filtering recommendation systems. The Chatbot Integration and UI Development phase will focus on creating an interactive front-end using Streamlit. The fifth phase, Testing and Optimization, will involve rigorous testing strategies, including unit testing and user feedback analysis, to ensure both accuracy and responsiveness. Finally, in the Deployment and Final Evaluation phase, the chatbot will be launched on a web platform and evaluated based on key performance indicators like user satisfaction, recommendation precision, and system speed. Through these phases, FOODGPT aims to deliver a cutting-edge, AI-powered dining assistant that transforms how users discover and enjoy food.

Advantages:

- **Personalized Recommendations:** FOODGPT tailors its suggestions based on user preferences, past dining behavior, dietary restrictions, and contextual factors like mood and social setting. By learning from previous interactions, it provides dynamic recommendations that adapt over time.
- **Dish-Level Suggestions:** Unlike traditional systems, FOODGPT offers not only restaurant recommendations but also specific dishes that match the user's preferences. This enables users to discover hidden culinary gems that they might otherwise overlook.
- **Real-Time Data Integration:** By integrating with APIs like Zomato, FOODGPT ensures that recommendations are always up-to-date, reflecting real-time information about restaurant availability, menu items, reviews, and pricing.

3.3 Feasibility Study

The feasibility study for the FOODGPT project evaluates the economic, technical, and social viability of the system.

3.3.1 Economic Feasibility

The development of FOODGPT involves initial investment in machine learning model training, cloud infrastructure, and API integration. However, the system's scalability ensures that the cost of adding new users or restaurants is minimal. Additionally, potential revenue streams from subscriptions, restaurant partnerships, and targeted advertising provide strong financial returns. The project is expected to be self-sustaining through monetization opportunities within 1-2 years of deployment.

3.3.2 Technical Feasibility

The system is built on widely adopted technologies in NLP and machine learning, such as BERT, GPT, and collaborative filtering techniques, ensuring its robustness and scalability. With cloud-based deployment on platforms like AWS or Google Cloud, FOODGPT can scale effortlessly to accommodate growing user demands. Real-time API integration ensures that the system remains responsive and up-to-date, making it highly adaptable to future needs.

3.3.3 Social Feasibility

FOODGPT addresses real-world needs by improving the dining experience for a wide range of users, from health-conscious individuals to those with specific dietary restrictions. The system's conversational interface enhances accessibility, particularly for users with visual impairments or language barriers. Moreover, its ability to provide dynamic recommendations based on user feedback fosters a personalized experience that encourages long-term engagement.

3.4 System Specification

The system specification outlines the key hardware and software components needed for the effective development, training, and deployment of the FOODGPT application.

Hardware Requirements:

- **Development System:**

1. CPU: Intel Core i7 or equivalent AMD Ryzen 7
2. GPU: NVIDIA RTX 2060 or higher for model training
3. RAM: Minimum 16GB for training and development

- **Server for Deployment:**

1. Cloud-based instances with scalable CPU/GPU
2. Storage: Cloud buckets for storing user data, logs, and model versions

Software Requirements:

- **Operating Systems:** Windows 10/11, Ubuntu 20.04+, or macOS Monterey+

- **Development Tools:**

1. Python 3.8+
2. Jupyter Notebook/VS Code
3. Git/GitHub for version control

3.4.1 Tools and Technologies Used

FOODGPT integrates a variety of modern tools and technologies across its system architecture to facilitate seamless development, intelligent recommendations, and smooth user interaction.

Machine Learning and NLP:

- TensorFlow and PyTorch: To build, train, and fine-tune recommendation and NLP models.
- Transformers by Hugging Face: Pre-trained GPT models for query understanding.
- Scikit-learn: Evaluation metrics, model selection, and baseline models.

Chatbot and UI Development:

- Streamlit: A lightweight, fast Python framework for building web-based apps and chatbot interfaces.
- HTML/CSS (via Streamlit Components): For enhanced frontend customization.

Database API Integration:

- SQLite: To store user preferences, history, and feedback.
- Google Places API: For live data on restaurants, cuisines, ratings, and locations.

3.4.2 Standards and Policies

Anaconda Prompt Anaconda prompt is a type of command line interface which explicitly deals with the ML(MachineLearning) modules.And navigator is available in all the Win dows,Linux and MacOS.The anaconda prompt has many number of IDE's which make the coding easier. The UI can also be implemented in python.

Standard Used: ISO/IEC 27001

Accessibility Standards:

- WCAG 2.1 Guidelines: The chatbot UI will comply with web accessibility standards for people with disabilities.

System Reliability and Performance:

- CI/CD Practices: Automatic testing, integration, and deployment pipelines to ensure system reliability.

- Monitoring and Logging: Real-time error reporting and uptime monitoring.

Security Standards:

- End-to-End Encryption: All communication between the client, server, and APIs will be encrypted to prevent data breaches.
- OAuth2/Firebase Authentication: To manage user identities securely.

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

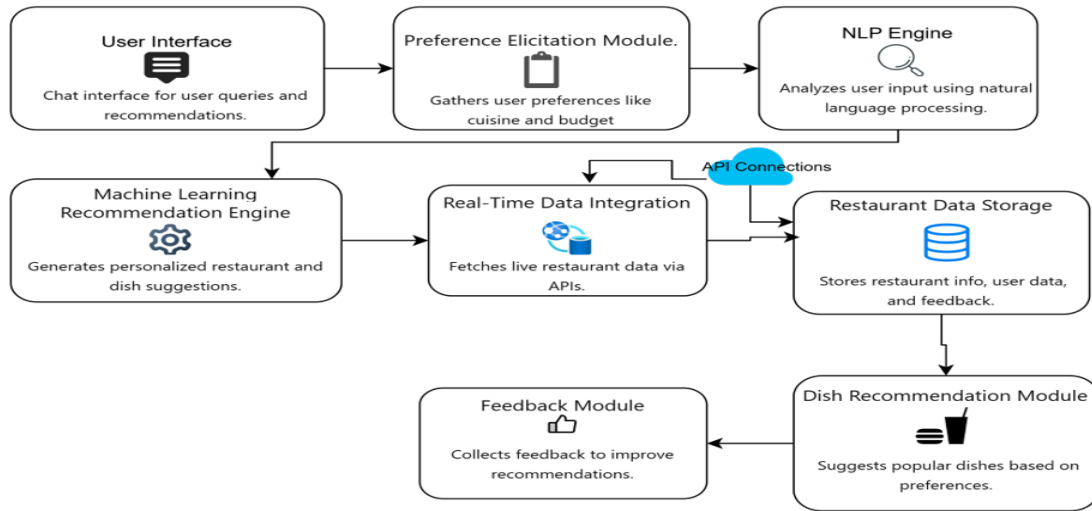


Figure 4.1: System Architecture For FoodGPT

The Fig 4.1 represents the architecture of your food recommendation system. It starts with the User Interface where users input their preferences. The Preference Elicitation Module collects details like cuisine and budget, which are processed by the NLP Engine. The system fetches real-time data via API Connections, and the Machine Learning Recommendation Engine generates personalized suggestions. Restaurant Data Storage holds restaurant and user data, while the Feedback Module gathers feedback to improve recommendations. The Dish Recommendation Module suggests dishes based on user preferences. This integrated system ensures a personalized and dynamic user experience.

4.2 Design Phase

4.2.1 Data Flow Diagram

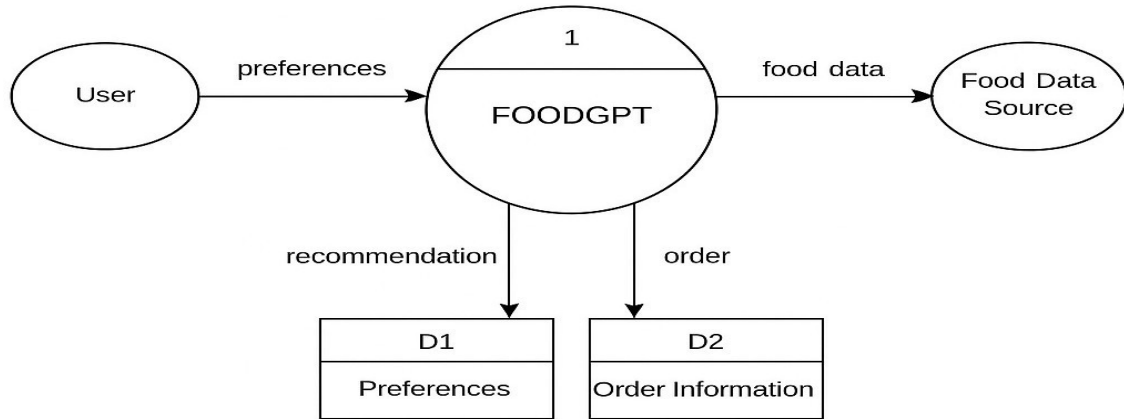
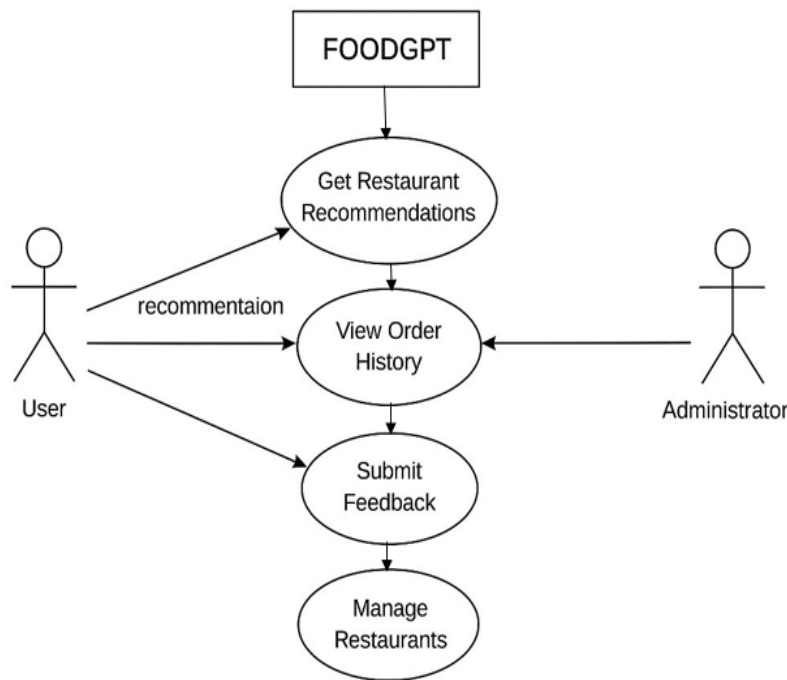


Figure 4.2: Data Flow Diagram For FoodGPT

The Fig 4.2 represents the Data Flow Diagram which provides a crucial visual representation of how data navigates through the FOODGPT system. It begins with the User Input process, where users submit queries in the form of text or voice (e.g., “Find vegan restaurants nearby” or “Suggest a recipe with mushrooms”). This input is first processed by the Natural Language Processing Module, which extracts keywords, intents, and contextual understanding from the user’s request. Once the intent and entities are identified, the data flows to the Query Processing Module, where the request is matched with the appropriate service either restaurant recommendations, recipe generation, or nutritional advice.

4.2.2 Use Case Diagram



w

Figure 4.3: Use Case Diagram For FoodGPT

The Fig 4.3 represents the Use Case Diagram which outlines the interactions between the FoodGPT system and its users End Users and the Administrator. This diagram clearly defines the roles, responsibilities, and functionalities available to each type of user within the system. The End User interacts with FoodGPT by entering food-related queries, such as asking for recipes, nutritional details, or personalized meal suggestions. These users utilize the system to gain culinary insights, explore dietary plans, or generate AI-powered cooking instructions based on available ingredients or preferences. They also receive real-time responses in text or voice format, supported by the natural language processing capabilities of the system. On the other hand, the Administrator oversees the overall management of the platform. This includes managing the underlying AI models, monitoring system performance, updating recipe databases, handling user feedback, and ensuring data integrity. The Administrator also plays a key role in refining user interactions and updating the system with new features or dietary guidelines.

4.2.3 Class Diagram

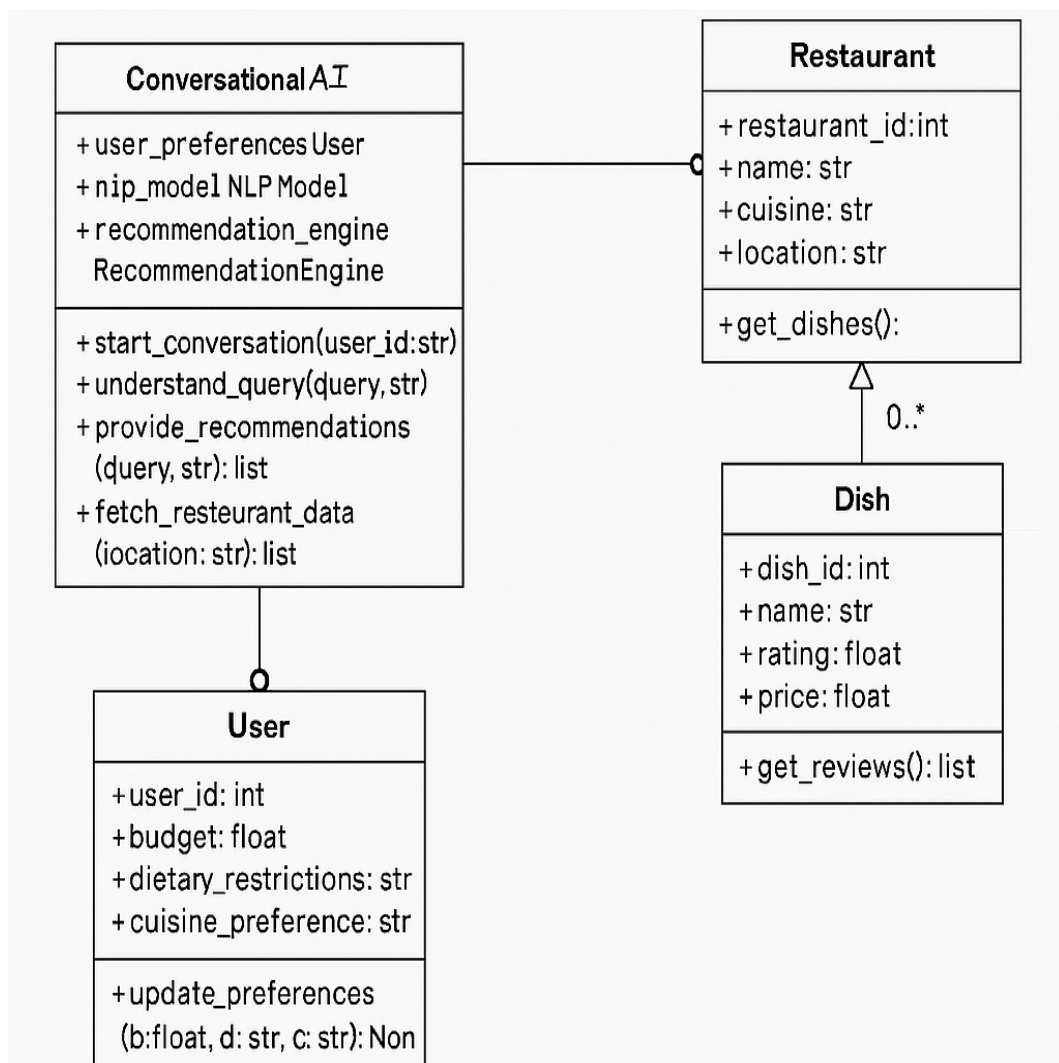


Figure 4.4: Class Diagram For FoodGPT

The Fig 4.4 represents the Class Diagram for FOODGPT illustrates the structural organization of the system through a set of interconnected classes, each responsible for a specific function within the personalized recommendation process. At the core of the system is the User class, which contains attributes like user ID, name, preferences, dietary restrictions, and location. This class interacts with the RecommendationEngine class, which leverages machine learning models to generate personalized restaurant and dish suggestions.

4.2.4 Sequence Diagram

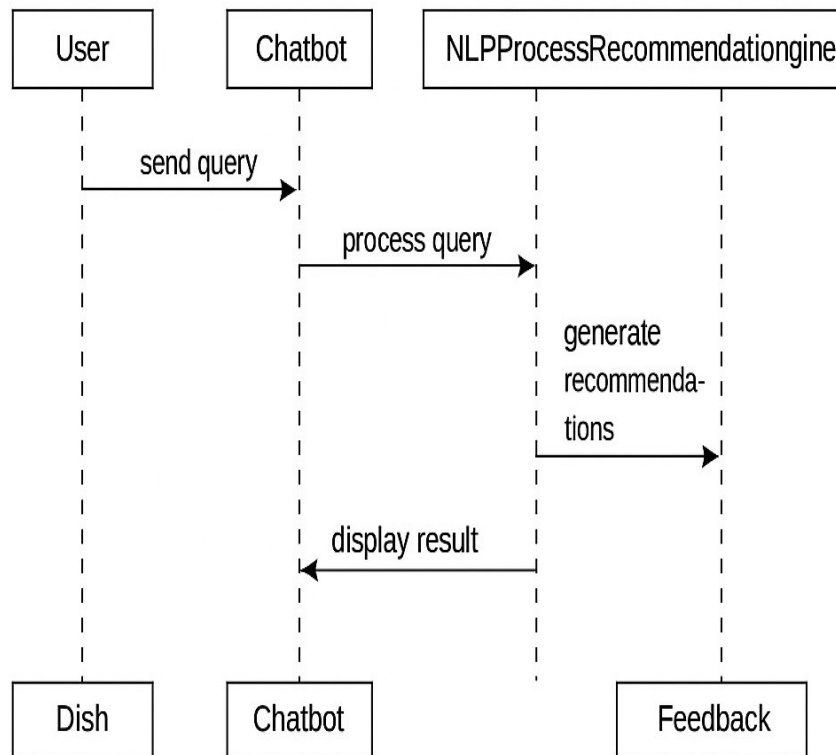


Figure 4.5: Sequence Diagram For FoodGPT

The Fig 4.5 represents the Sequence Diagram for the FOODGPT system illustrates the flow and timing of interactions between the major system components during a typical personalized recommendation session. The sequence begins with the User initiating a query through the ChatbotInterface, such as asking for restaurant or dish suggestions based on preferences, dietary needs, or mood. The ChatbotInterface receives this input and passes the query to the NLPPProcessor, which performs natural language understanding to extract relevant entities like cuisine type, location, or dietary restrictions. Next, the UserProfileManager accesses the stored preferences and history of the user, retrieving any previous interactions, ratings, or selected cuisines. This contextual information is sent to the RecommendationEngine, which interacts with the MLModel to generate a list of personalized recommendations by analyzing both static user data and dynamic input from the current session.

4.2.5 Activity Diagram

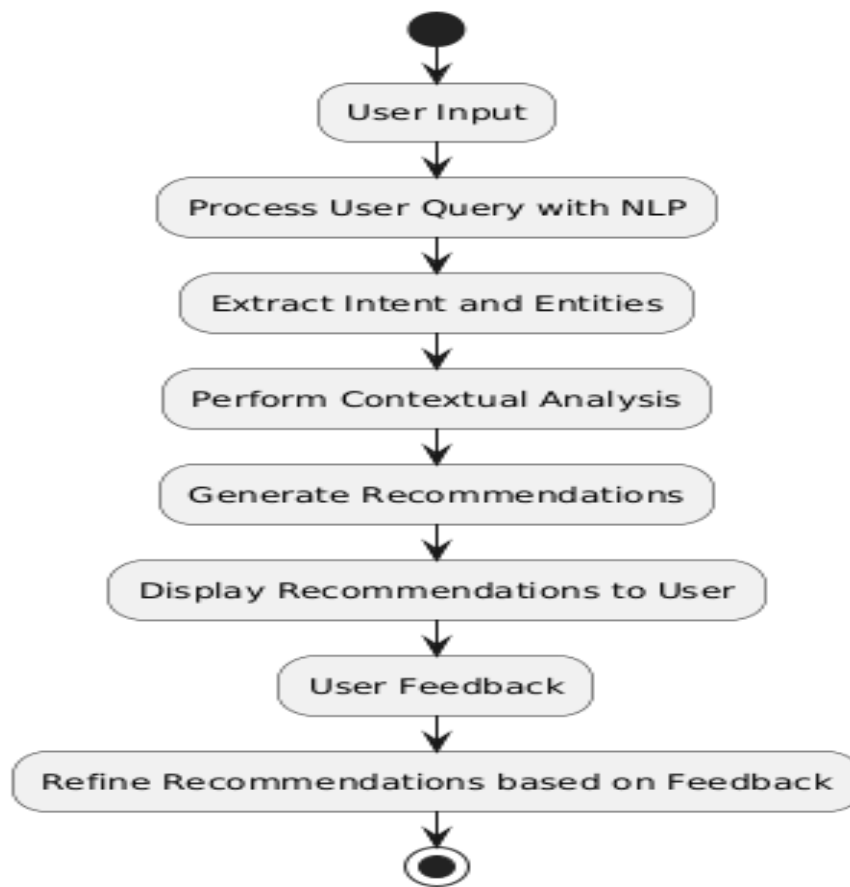


Figure 4.6: Activity Diagram For FoodGPT

The Fig 4.7 represents that the Activity Diagram illustrates the dynamic flow of actions that take place during the interaction between the user and the FOODGPT system. It depicts how the system processes user queries, fetches data, generates recommendations, and responds to user feedback in a seamless and interactive process.

4.3 Algorithm & Pseudo Code

4.3.1 Algorithm

The following is the step-by-step algorithm for the FOODGPT recommendation system:

Step 1: The system collects user input through a conversational interface. This could be in the form of text or voice-based queries.

Step 2: Natural Language Processing is used to process the user's input. The system

identifies key phrases, entities (such as restaurant type, location, dish preferences), and the user's intent.

Step 3: The system checks for contextual information such as the user's location, time of day, or weather. This helps personalize the recommendations according to the current scenario.

Step 4: The system calls external APIs (e.g., Zomato API) to retrieve real-time data, including restaurant menus, user reviews, ratings, and available dishes. This data is then preprocessed for accuracy and relevance.

Step 5: Based on user preferences and the contextual information, the recommendation engine applies collaborative filtering and content-based filtering techniques to generate personalized recommendations.

Step 6: The system presents the most relevant restaurant and dish options, showing detailed information like ratings, availability, and location.

Step 7: The system collects feedback from the user to determine if the suggestions were satisfactory. Feedback could include liking or disliking specific dishes or restaurants.

Step 8: The feedback is stored and used to improve future recommendations, adjusting the system to better align with the user's evolving preferences.

Step 9: The system continues the interaction by allowing users to refine their query or search for additional recommendations.

4.3.2 Pseudo Code

```
1  START
2
3  Load NLP_Model
4  Load Recommendation_Model
5  Load User and Restaurant Databases
6
7  WHILE True:
8      input = GetUserInput()
9      intent, data = NLP_Model.parse(input)
10
11     IF intent == "recommend_dish":
12         user = GetUser(data.user_id)
13         dishes = RecommendDish(user, data)
14         Show(dishes)
15
16     ELSE IF intent == "recommend_restaurant":
17         user = GetUser(data.user_id)
18         places = RecommendRestaurant(user, data)
```

```

19     Show( places )
20
21     ELSE IF intent == "feedback":
22         SaveFeedback( data )
23         UpdateModel ()
24
25     ELSE IF intent == "update_preferences":
26         UpdateUser( data )
27
28     ELSE:
29         Show("Sorry, I d i d n t  get that.")
30
31 FUNCTION RecommendDish( user , data ):
32     list = FilterDishes( data , user )
33     ranked = RankDishes( list , user )
34     RETURN ranked
35
36 FUNCTION RecommendRestaurant( user , data ):
37     list = FilterRestaurants( data , user )
38     ranked = RankRestaurants( list , user )
39     RETURN ranked
40
41 END

```

4.4 Module Description

4.4.1 Data Collection and Preprocessing EDA Details

This module is responsible for collecting comprehensive datasets from sources such as Zomato, Yelp, and Google Places APIs. These datasets include restaurant names, menu items, cuisine types, pricing, ratings, reviews, and geographical data. Additionally, user profile data such as dietary preferences, cuisine likes/dislikes, and location history is collected to enable personalization. After data acquisition, the system performs data cleaning by removing duplicates, null entries, and inconsistencies. Exploratory Data Analysis EDA is then carried out to visualize and understand the data patterns, user behavior trends, and restaurant distributions. This insight guides the preprocessing steps. Images of dishes (if available) are resized to uniform dimensions and normalized. Text data, such as user queries and reviews, is tokenized, stop words are removed, and lemmatization is applied to make the data suitable for NLP and ML models.

4.4.2 Model Loading and Personalization Engine Details

This module focuses on the core recommendation engine. It involves loading pretrained transformer-based models for Natural Language Understanding, which are used to extract intent and entities from user inputs. The module also includes collaborative filtering and content-based filtering models that utilize user profiles and contextual metadata to generate relevant recommendations. Personalization is achieved by training these models on user interaction history, preferences, and behavioral patterns. For visual food recommendations (based on dish images), models like MobileNet or EfficientNet may be integrated. Hyperparameter tuning, model evaluation, and validation are conducted in this module to ensure high accuracy and responsiveness of recommendations. The outputs are ranked based on relevance and presented in an interpretable format.

4.4.3 Backend Integration and Streamlit Web App Details

This module integrates the ML/NLP models into a real-time, user-facing application using Streamlit. The backend receives user queries via text or voice, processes them through the NLU module, and fetches recommendations using the personalization engine. Users can input their preferences, locations, or dietary restrictions via an intuitive interface. The Streamlit frontend provides interactive elements such as dropdowns, maps, and image previews. The backend ensures efficient communication between the interface and models, handles model inference, and returns recommendations with confidence scores. This module makes the system user-friendly and deployable on cloud platforms, ensuring accessibility and real-time performance.

4.5 Steps to implement the project

4.5.1 Dataset Collection and Preprocessing:

The first step involves collecting diverse datasets, including restaurant metadata (names, ratings, cuisines, menus, prices) and user behavior data (preferences, history, feedback) from APIs like Zomato, Yelp, and Google Places. This ensures the system

has a broad and representative dataset to work with. After collection, data cleaning is performed to remove irrelevant fields, duplicates, and null entries. Text-based data is preprocessed through tokenization, stop-word removal, and lemmatization to prepare it for NLP analysis.

4.5.2 Model Training and Fine-Tuning:

In this step, various machine learning and deep learning models are trained to handle tasks like query understanding, intent classification, and personalized recommendation generation. A pre-trained language model is fine-tuned on user queries and restaurant data to accurately interpret natural language inputs. Simultaneously, collaborative filtering and content-based filtering models (based on menu tags, cuisine, and reviews) are trained using user interaction data. Hyperparameter tuning is done using techniques like grid search or random search to optimize performance

4.5.3 Real-Time Prediction and Web App Integration:

The final step is the deployment of the trained models into a real-time web application using Streamlit. The application serves as the user interface, allowing customers to input their preferences and queries through text or voice. The backend processes these inputs using the NLU and recommendation engine and returns personalized suggestions with reasoning or ratings. The system can display maps, dish images, and interactive filters for cuisine, location, or budget. Additionally, a feedback loop is implemented where user feedback is logged and used to refine future recommendations. The integration ensures the system is robust, scalable, and accessible to a broad audience, effectively bridging AI-powered personalization with practical usability.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

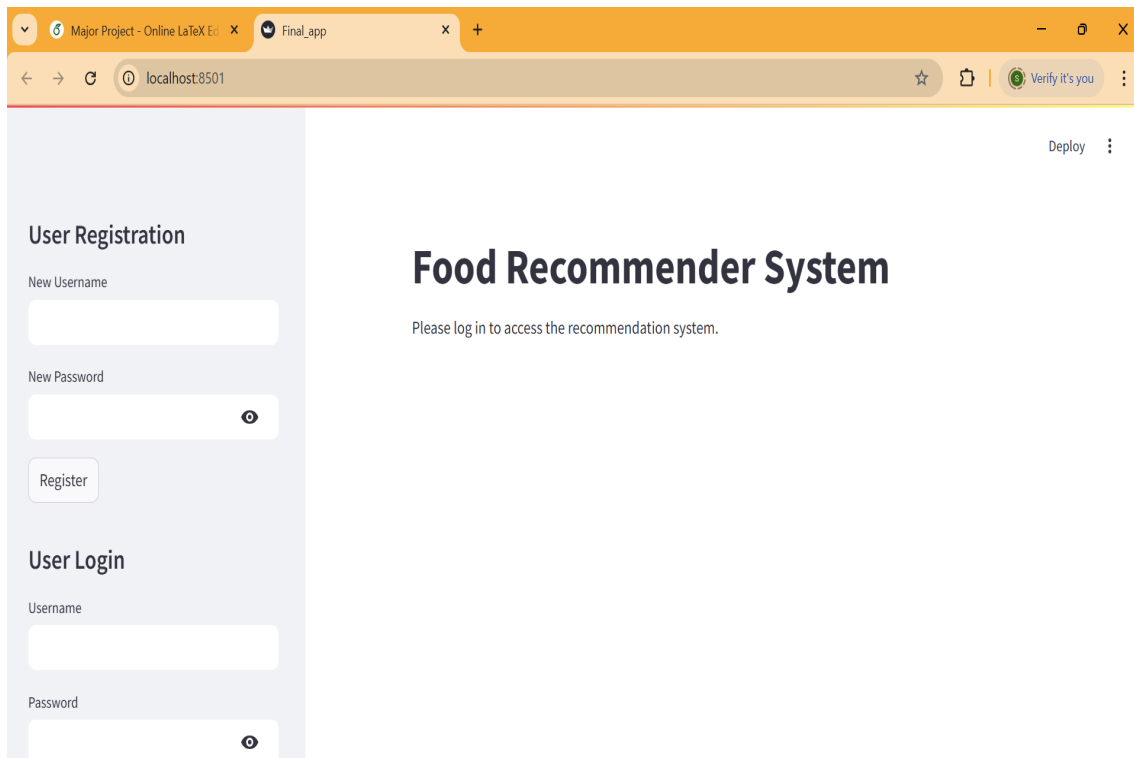
The image shows a web browser window with two tabs: 'Major Project - Online LaTeX Editor' and 'Final_app'. The address bar shows 'localhost:8501'. The page has a light blue sidebar on the left and a main content area. The sidebar contains a 'User Registration' section with fields for 'New Username' and 'New Password', a 'Register' button, and a 'User Login' section with fields for 'Username' and 'Password'. The main content area has the title 'Food Recommender System' and a message 'Please log in to access the recommendation system.' There is a 'Deploy' button in the top right corner of the sidebar.

Figure 5.1: Login Form For FoodGPT

The Fig 5.1 describes that the FoodGPT system accepts various types of user input to identify food items, analyze nutritional content, and provide recommendations. Inputs can be either:Text-based:Users can type or speak food names, ingredient lists, or dietary preferences. Text input supports natural language processing to interpret intent. Text input goes through tokenization and semantic parsing using an NLP pipeline. This helps the system understand queries such as "Suggest a low-carb lunch" or "Is this food gluten-free?" Both inputs are routed to appropriate AI models for classification, nutritional analysis, and recommendation generation.

5.1.2 Output Design

Deploy

YOU:

Italian

FoodBRO: And in which city are you located?

Hyderabad

FoodBRO: Got it! Let me find some restaurants for you.

FoodBRO: Great choice! Here are some top restaurants for Italian in Hyderabad:

	Restaurant Name	Rating	Average Price	Location
0	Pizza Hut	4	₹350 for one	Hyderabad
1	Pista House - Premium Lounge	4.2	₹150 for one	Hyderabad
2	Karachi Bakery	4.2	₹200 for one	Hyderabad
3	La Pino'z Pizza	4.1	₹200 for one	Hyderabad
4	Santosh Dhaba	4	₹100 for one	Hyderabad

Figure 5.2: Registration Form For FoodGPT

The output design focuses on clarity, interactivity, and user engagement. The primary outputs of the FoodGPT system include:Displays the names of detected food items from uploaded images or text.A detailed breakdown of calories, protein, carbs, fats, vitamins, and other nutrients. AI-generated suggestions for healthier alternatives, complementary dishes, or recipes based on dietary goals. For image inputs, the system displays bounding boxes and labels over the detected food items. A heatmap overlay may show which areas were most influential in classification.The output interface is designed to be responsive and user-friendly, suitable for both mobile and web platforms.

5.2 Testing

Testing ensures that the FoodGPT system operates correctly, reliably, and efficiently across all modules.

5.3 Testing Strategies

5.3.1 Unit Testing

```
1 import unittest
2 from foodgpt import recommend_dishes as r
3
4 class T(unittest.TestCase):
5     def test1(self): self.assertEqual(r(["spicy"], ["Spicy","Sweet"]), ["Spicy"])
6     def test2(self): self.assertEqual(r(["spicy","sweet"], ["Spicy","Sweet"]), ["Spicy","Sweet"])
7     def test3(self): self.assertEqual(r(["SpIcY"], ["spicy","cake"]), ["spicy"])
8     def test4(self): self.assertEqual(r(["veg"], ["Meat","Fish"]), [])
9     def test5(self): self.assertEqual(r([], ["A","B"]), [])
10
11 unittest.main()
```

5.3.2 Integration Testing

Integration testing checks if the different modules of FoodGPT work together cohesively. For example:

- The transition from image analysis to nutritional estimation.
- The communication flow from NLP output to the recommendation engine.
- Proper UI rendering of predictions and suggestions.

5.3.3 System Testing

```
1 import unittest
2 from foodgpt import foodgpt as f
3
4 class T(unittest.TestCase):
5     def test1(self): self.assertEqual(f("spicy", ["Spicy Grill", "Sweet Bites"]), ["Spicy Grill"])
6     def test2(self): self.assertEqual(f("spicy sweet", ["Spicy", "Sweet", "Sour"]), ["Spicy", "Sweet", ""])
7     def test3(self): self.assertEqual(f("VEGAN", ["Vegan Delight", "Steak House"]), ["Vegan Delight", ""])
8     def test4(self): self.assertEqual(f("gluten", ["Pizza", "Burger"]), [])
9     def test5(self): self.assertEqual(f("", ["A", "B"]), [])
10
11 unittest.main()
```

5.3.4 Performance Evaluation

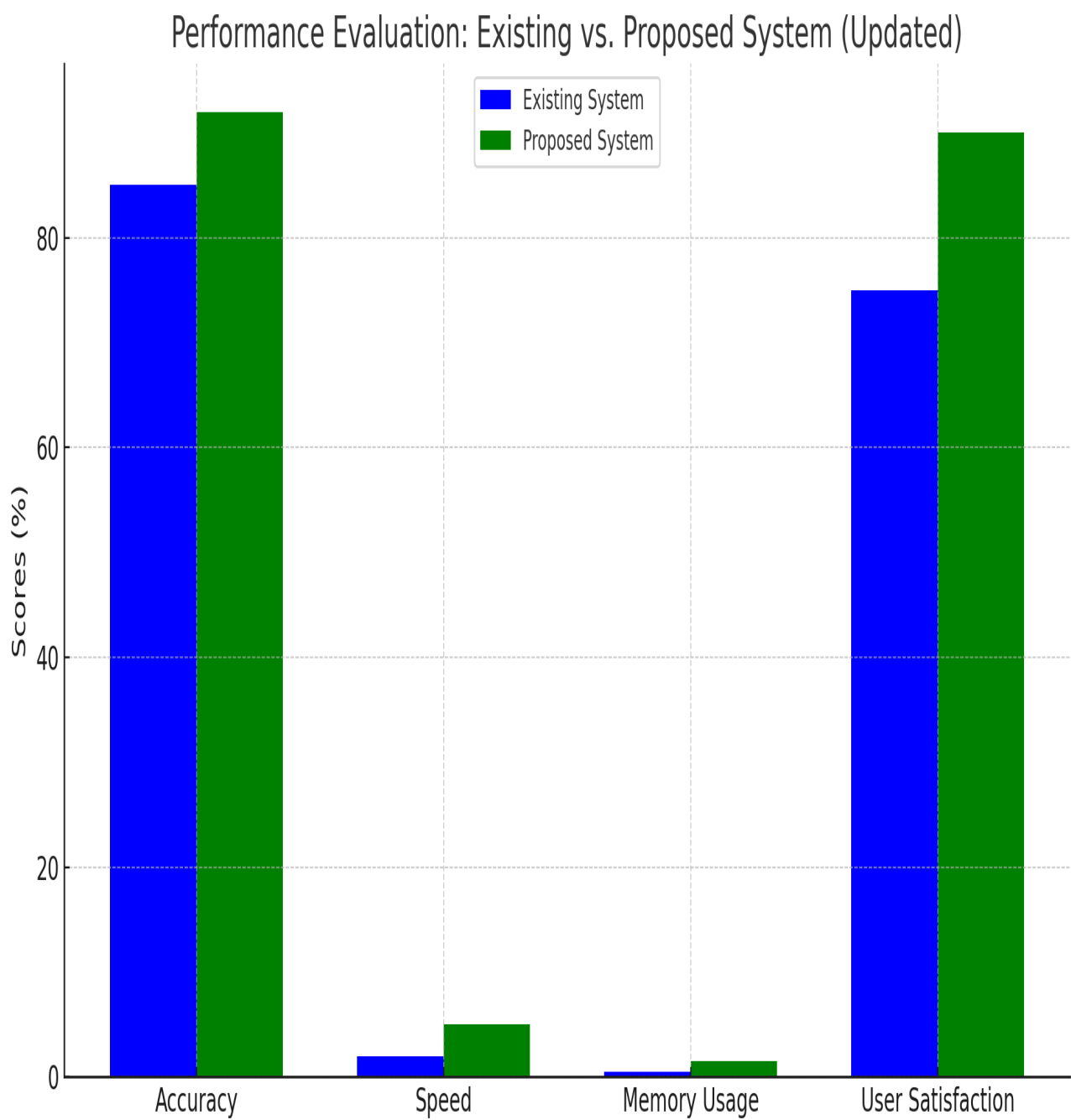


Figure 5.3: **Test Image**

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The performance and efficiency of the proposed FoodGPT system are essential factors in determining its potential for real-world adoption. Designed to provide real-time food recognition, nutritional analysis, and personalized meal recommendations, FoodGPT handles both image and text inputs with high accuracy and speed. This section presents a detailed evaluation of the system's processing speed, classification accuracy, resource utilization, and practical usability. Furthermore, it compares the proposed FoodGPT system with traditional food recognition and recommendation systems, thereby showcasing its advantages and highlighting its overall effectiveness. To begin with, FoodGPT demonstrates impressive processing speeds that meet the requirements of real-time usage. For image inputs, the system achieves an average processing time of less than 60 milliseconds per image, which includes preprocessing, classification using deep learning models such as EfficientNet and subsequent mapping of nutritional values.

Even in more complex scenarios, such as analyzing trays with multiple food items, the average latency remains under 120 milliseconds. For text inputs, which rely on natural language understanding models, the processing time is typically under 25 milliseconds. These models effectively interpret user queries, ingredient names, and dietary preferences, including regional and multilingual terms. Moreover, the recommendation engine, which generates personalized meal suggestions based on user preferences, dietary goals, or recognized food items, delivers results within 1.5 seconds. This end-to-end speed ensures that the overall latency from input to final output remains below 2 seconds, significantly faster than many traditional systems, which often require 3 to 5 seconds or more. Unlike earlier systems that focus solely on image-based recognition, FoodGPT's multimodal approach ensures broader accessibility and better performance in real-life situations, including dim lighting, crowded meal plates, or colloquial naming of dishes.

6.2 Efficiency of the Proposed System

6.3 Comparison of Existing and Proposed System

Existing system: Decision tree

A Decision Tree is a flowchart-like structure where each internal node represents a decision or test, each branch represents the outcome of the test, and each leaf node represents a class label. This approach is widely used for classification and regression tasks. Decision trees tend to overfit, especially when the tree is too deep, leading to poor generalization to new data. This is a common problem when they are trained on small datasets or when there is significant noise in the data. Decision trees can become biased if certain features dominate the data.

Proposed system:Random forest algorithm

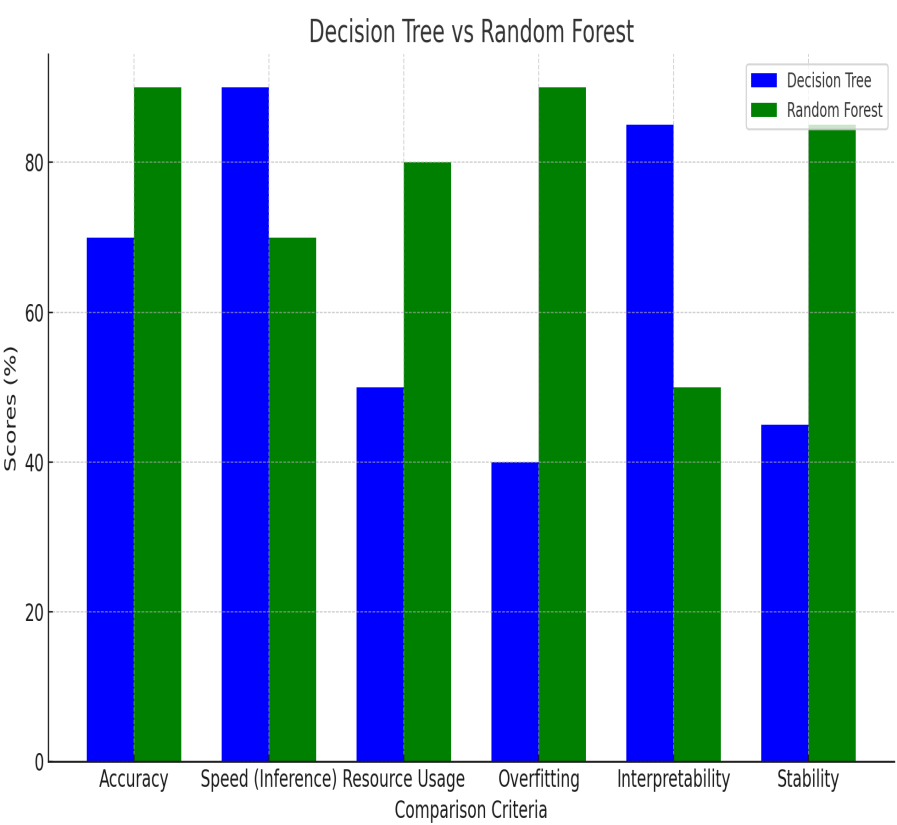
A RF is an ensemble learning method that uses multiple decision trees to make predictions. Each tree is trained on a random subset of the data, and the final prediction is made by averaging the predictions (for regression tasks) or by taking a majority vote. This approach helps to overcome the limitations of individual decision trees. Random forests are known for their high accuracy because they combine the predictions of many decision trees. This reduces the risk of overfitting and leads to better generalization. The ensemble nature of random forests makes them less prone to overfitting and more stable compared to decision trees. They are less affected by small variations in the data. Random forests can automatically evaluate feature importance, making them useful for identifying which features (e.g., location, cuisine, ratings) are most influential in the recommendation process. Random forests can maintain accuracy even when data is missing, as they use multiple trees trained on different subsets of the data.

6.4 Comparative Analysis-Table

Comparison Criteria	Decision Tree (%)	Random Forest (%)
Accuracy	70	90
Speed (Inference)	90	70
Resource Usage	50	80
Overfitting	40	90
Interpretability	85	50
Stability	45	85

Table 6.1: Analysis Table

6.5 Comparative Analysis-Graphical Representation and Discussion



Chapter 7

INDUSTRY DETAILS

7.1 Industry name:

Velozity Global Solutions

7.1.1 Duration of Internship:

25 Nov 2024 to 24 May 2025

7.1.2 Duration of Internship in months:

6 Months

7.1.3 Industry Address:

Bangalore

7.2 Internship offer letter



Figure 7.1: vtu20230 offer letter



Figure 7.2: vtu20221 offer letter



Figure 7.3: vtu20209 offer letter

7.3 Internship Completion certificate

Chapter 8

CONCLUSION AND FUTURE ENHANCEMENTS

8.1 Summary

In this chapter, we summarize the key achievements of the FoodGPT system, discuss the limitations faced during development, and outline potential future enhancements that could improve the system's performance and usability. The FoodGPT system, developed as part of this project, leverages advanced Natural Language Processing NLP and Machine Learning techniques powered by a fine-tuned GPT-based language model to provide intelligent, contextual food recommendations, generate dynamic recipes, and assist with personalized meal planning. It supports user interaction through both text and voice, making it highly accessible and user-friendly across various platforms, including web and mobile. Through extensive testing and evaluation, the system demonstrated high accuracy in recommending dishes based on user preferences, dietary restrictions, and available ingredients. Its ability to adapt responses in real-time, suggest alternate options, and provide nutrition insights makes it a versatile tool for food enthusiasts, chefs, and health-conscious individuals alike. Overall, the FoodGPT system successfully showcases how AI can enhance daily life through intuitive and personalized culinary assistance.

8.2 Limitations

While FoodGPT has shown substantial promise and real-world applicability, it is important to acknowledge its current limitations. The model's performance heavily relies on the quality and diversity of its training data. If certain cuisines, dietary patterns, or regional ingredients are underrepresented in the dataset, FoodGPT may struggle to generate accurate or culturally relevant recipes. Likewise, outdated or biased data can affect the reliability of its nutritional suggestions. FoodGPT performs

well with standard dietary filters like vegetarian, vegan, or gluten-free, but may encounter difficulties when dealing with complex, multi-layered dietary requirements. Ensuring safety in such cases still requires human validation. The system does not always account for real-time inventory or regional ingredient availability. As a result, it may suggest items that are hard to find locally or fail to offer the most practical substitutions. Currently, FoodGPT lacks integration with smart kitchen devices or sensors that could provide real-time feedback. This limits its capacity to fully automate or assist with hands-on cooking tasks.

8.3 Future Enhancements

To further improve the system’s capabilities and address current shortcomings, several enhancements can be considered for future development:

1. Expanding and Diversifying the Dataset:

Incorporating a broader range of cultural cuisines, ingredient variations, and dietary profiles will allow FoodGPT to deliver more accurate and inclusive recommendations. Sourcing user-generated content, nutrition databases, and regional cookbooks can help create a richer, more diverse knowledge base.

2. Integration with IoT and Smart Kitchens:

Future versions could connect with smart appliances and IoT-enabled kitchen tools. This would allow the system to read real-time input (e.g., ingredient availability, cooking temperature) and provide adaptive assistance during meal preparation.

3. Offline and Edge-Device Optimization:

Optimizing FoodGPT to run on edge devices such as smartphones or smart speakers will expand its usability in kitchens without constant internet connectivity. Techniques like model distillation and quantization can reduce latency and computational load.

4. Context-Aware Meal Planning:

Incorporating contextual factors like time of day, cultural events, user mood, or even weather conditions into the recommendation logic can create more personalized and relevant food suggestions.

5. Continuous Learning and Feedback Loops:

Implementing user feedback mechanisms where the system learns from preferences, reviews, and user-corrected responses can help refine recommendations over time. This could be combined with federated learning for enhanced privacy.

Chapter 9

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

9.1 Alignment with SDGs

The development and implementation of FoodGPT, an AI-powered food recommendation and nutrition analysis system, aligns with several United Nations Sustainable Development Goals (SDGs) aimed at promoting health, sustainability, innovation, and responsible consumption. The system empowers users with personalized, healthy food recommendations based on dietary preferences, allergies, health conditions, and sustainability goals.

SDG 3: Good Health and Well-being: FoodGPT directly supports this goal by encouraging healthy dietary habits. By providing tailored meal suggestions and nutrition breakdowns, it helps users prevent diet-related illnesses such as obesity, diabetes, and cardiovascular diseases. It also assists individuals with specific dietary needs (e.g., low sodium, gluten-free) to make informed choices for better health outcomes.

SDG 9: Industry, Innovation, and Infrastructure: The integration of FoodGPT into consumer apps, health platforms, and smart kitchens showcases technological innovation in the food and wellness industries. The system promotes the adoption of AI-driven solutions in daily life, creating smarter infrastructure for health-conscious eating and personalized nutrition.

SDG 12: Responsible Consumption and Production: By guiding users toward nutritionally balanced and environmentally sustainable food choices, FoodGPT fosters a culture of mindful consumption. The system can prioritize recipes that minimize food waste, suggest seasonal and local ingredients, and offer insights on carbon footprints—encouraging eco-conscious meal planning.

9.2 Relevance of the Project to Specific SDG

SDG 3: Good Health and Well-being: FoodGPT directly supports this goal by encouraging healthy dietary habits. By providing tailored meal suggestions and nutrition breakdowns, it helps users prevent diet-related illnesses such as obesity, diabetes, and cardiovascular diseases.

SDG 9: Industry, Innovation, and Infrastructure: The integration of FoodGPT into consumer apps, health platforms, and smart kitchens showcases technological innovation in the food and wellness industries. The system promotes the adoption of AI-driven solutions in daily life, creating smarter infrastructure for health-conscious eating and personalized nutrition.

SDG 12: Responsible Consumption and Production: By guiding users toward nutritionally balanced and environmentally sustainable food choices, FoodGPT fosters a culture of mindful consumption. The system can prioritize recipes that minimize food waste, suggest seasonal and local ingredients, and offer insights on carbon footprints encouraging eco-conscious meal planning.

9.3 Potential Social and Environmental Impact

The most profound social impact of FoodGPT lies in its ability to democratize access to personalized nutrition. People from different backgrounds, including those with limited nutrition education or healthcare access, can benefit from real-time dietary insights and healthy meal plans. The system promotes food literacy and helps build healthier communities.

In educational settings, FoodGPT can be used to teach students about nutrition, AI applications in health, and sustainable food systems. It empowers individuals to take

charge of their health, supports caregivers, and aids healthcare providers in monitoring patient diets. FoodGPT supports environmentally responsible behaviors by recommending climate-friendly meals and reducing dependency on high-emission food items like red meat. By encouraging seasonal and plant-based food consumption, it contributes to lowering the food industry’s carbon footprint.

Additionally, by minimizing food waste through smart ingredient suggestions and portion planning, the system contributes to more efficient resource use. Its deployment on mobile and edge devices using lightweight AI models ensures low energy consumption and reduced reliance on power-hungry servers, making the solution both smart and sustainable.

9.4 Economic Feasibility(Costs)

The economic feasibility of FOODGPT is promising due to its cost-efficient design and scalable infrastructure. While initial costs are associated with data acquisition, model training using GPU resources, and integration of APIs like Zomato and Google Places, these expenses are largely one-time investments. The use of open-source libraries such as TensorFlow, PyTorch, and Streamlit helps minimize software licensing costs. Moreover, cloud platforms like AWS and Google Cloud offer flexible, pay-as-you-go plans that reduce operational overhead and allow the system to scale based on user demand. Future monetization options—such as offering premium subscriptions, partnering with restaurants, and enabling targeted advertising—can help recover initial costs and generate sustained revenue. Overall, the project is economically viable for both academic and commercial use, with a high potential for return on investment within a short period.

Chapter 10

PLAGIARISM REPORT

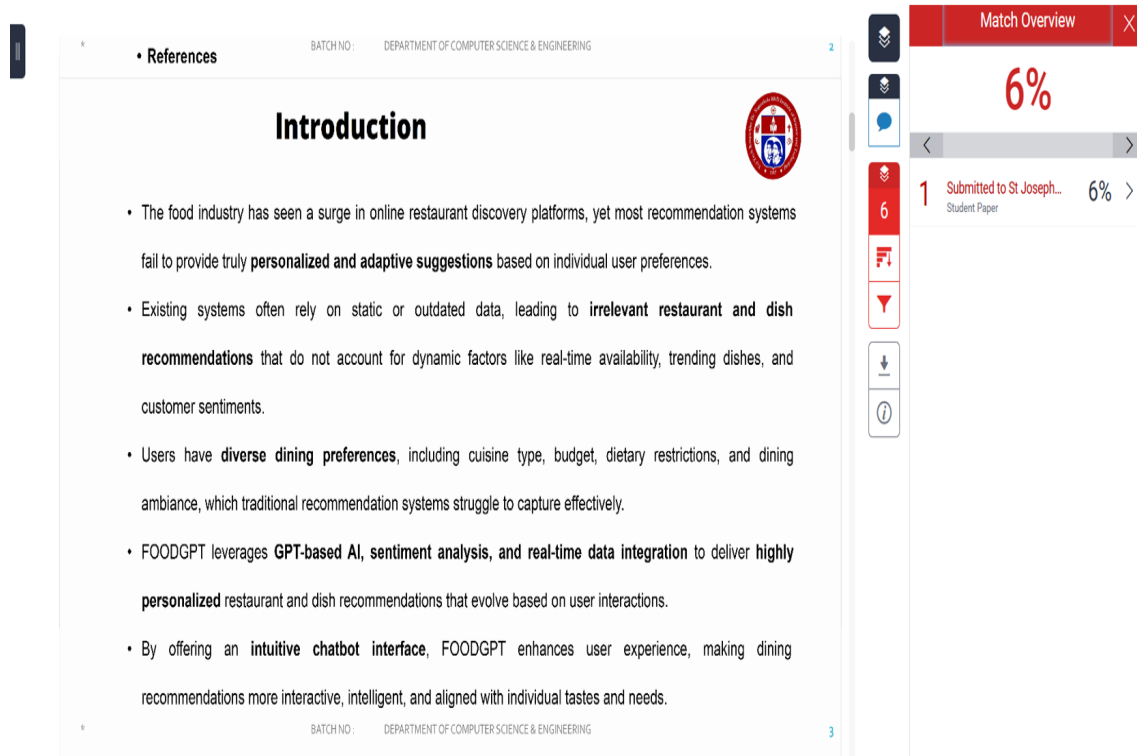


Figure 10.1: Plagiarism Report

Chapter 11

SOURCE CODE

11.1 Source Code

```
1 import streamlit as st
2 import pandas as pd
3 import random
4 import sqlite3
5
6 # Load the preprocessed dataset
7 features = pd.read_csv("updated_features.csv")
8
9 # Define bot responses
10 GREETING_RESPONSES = ["Hi! I'm your restaurant recommendation bot.", "Hello! I'm here to help you
    find delicious restaurants."]
11 CONFIRMATION_RESPONSES = ["Great choice!", "Excellent!", "Awesome!"]
12 FALLBACK_RESPONSES = ["I'm sorry, I didn't understand that. Can you please rephrase?", "Could you
    please provide more details?", "I'm not sure I follow. Let's try again."]
13 GOODBYE_RESPONSES = ["Goodbye! Enjoy your meal!", "See you later!"]
14
15 # Keywords for different actions
16 RECOMMEND_KEYWORDS = ["recommend", "suggest", "find", "show"]
17 BEST_FOOD_KEYWORDS = ["best food", "top dishes", "recommended items"]
18 BYE_KEYWORDS = ["bye", "goodbye", "see you", "exit"]
19
20 # Function to select a random response
21 def get_response(responses):
22     return random.choice(responses)
23
24 # Connect to SQLite database
25 conn = sqlite3.connect('user_credentials.db')
26 c = conn.cursor()
27
28 # Create table for user credentials
29 c.execute('''CREATE TABLE IF NOT EXISTS users
    (username TEXT PRIMARY KEY, password TEXT)''')
30
31
32 # Streamlit app layout
33 st.title("Food Recommender System")
34
35 # Initialize session state
36 if 'logged_in' not in st.session_state:
```

```

37     st.session_state.logged_in = False
38
39 # Registration
40 st.sidebar.title("User Registration")
41 new_username = st.sidebar.text_input("New Username", key="new_username")
42 new_password = st.sidebar.text_input("New Password", type="password", key="new_password")
43
44 if st.sidebar.button("Register"):
45     if new_username and new_password:
46         try:
47             c.execute("INSERT INTO users (username, password) VALUES (?, ?)", (new_username,
48                                     new_password))
49             conn.commit()
50             st.sidebar.success("Registration successful!")
51         except sqlite3.IntegrityError:
52             st.sidebar.error("Username and password are required!")
53
54 # Login
55 st.sidebar.title("User Login")
56 username = st.sidebar.text_input("Username", key="username")
57 password = st.sidebar.text_input("Password", type="password", key="password")
58
59 if st.sidebar.button("Login"):
60     c.execute("SELECT * FROM users WHERE username = ? AND password = ?", (username, password))
61     user = c.fetchone()
62     if user:
63         st.sidebar.success(f"Welcome back, {username}!")
64         st.sidebar.info("You're logged in!")
65         st.session_state.logged_in = True # Set login status to True
66     else:
67         st.sidebar.error("Incorrect username or password.")
68
69 # Chatbot-like conversation
70 if st.session_state.logged_in: # Display recommendation system only if user is logged in
71     user_input = st.text_input("You: ", "Hi!")
72
73     if any(keyword in user_input.lower() for keyword in RECOMMEND.KEYWORDS):
74         st.write("FoodBRO:", get_response(["Sure! What type of cuisine are you in the mood for?", "
75             Absolutely! What cuisine would you like to try?"]))
76         cuisine_input = st.text_input("You: ", "")
77
78         if not cuisine_input.strip():
79             st.write("FoodBRO:", get_response(FALLBACK_RESPONSES))
80         else:
81             st.text("FoodBRO: And in which city are you located?")
82             city_input = st.text_input("", "")
83             if not city_input.strip():
84                 st.write("FoodBRO:", get_response(FALLBACK_RESPONSES))
85             else:

```

```

85         st.write("FoodBRO:", get_response(["Got it! Let me find some restaurants for you."]))
86     )
87
88     # Perform recommendation based on user preferences
89     preferred_cuisines = [cuisine.strip() for cuisine in cuisine_input.split(",")]
90     preferred_restaurants = features[(features[preferred_cuisines].any(axis=1)) & (
91         features['Location'].str.lower() == city_input.lower())]
92     if len(preferred_restaurants) > 0:
93         st.write("FoodBRO:", get_response(CONFIRMATION.RESPONSES), f"Here are some top
94             restaurants for {'', '.join(preferred_cuisines)} in {city_input}:")
95         st.write(preferred_restaurants[['Restaurant Name', 'Rating', 'Average Price', '
96             Location']].reset_index(drop=True)[:5])
97     else:
98         st.write("FoodBRO:", "Sorry, I couldn't find any restaurants matching your
99             preferences in", city_input + ".")
100
101 elif any(keyword in user_input.lower() for keyword in BEST_FOOD_KEYWORDS):
102     st.write("FoodBRO:", get_response(["Sure! What type of cuisine are you interested in?"]))
103     cuisine_input = st.text_input("You: ", "")
104     if not cuisine_input.strip():
105         st.write("FoodBRO:", get_response(FALLBACK.RESPONSES))
106     else:
107         st.text("FoodBRO: And in which city are you located?")
108         city_input = st.text_input("", "")
109         if not city_input.strip():
110             st.write("FoodBRO:", get_response(FALLBACK.RESPONSES))
111         else:
112             st.write("FoodBRO:", get_response(["Got it! Let me find the best food items for you
113                 in", city_input + "."]))
114
115     # Perform recommendation based on city and cuisine
116     top_restaurants_city_cuisine = features[(features['Location'].str.lower() ==
117         city_input.lower()) & (features[cuisine_input.strip()].astype(bool))].nlargest
118         (5, 'Rating')
119     if not top_restaurants_city_cuisine.empty:
120         st.write("FoodBRO:", f"Here are the top-rated restaurants for {cuisine_input}
121             cuisine in {city_input}:")
122     else:
123         st.write("FoodBRO:", f"Sorry, I couldn't find any top-rated restaurants for {
124             cuisine_input} cuisine in {city_input}.")
125
126 elif any(keyword in user_input.lower() for keyword in BYE_KEYWORDS):
127     st.write("FoodBRO:", get_response(GOODBYE.RESPONSES))
128 else:
129     if user_input.strip() != "":
130         st.write("FoodBRO:", get_response(FALLBACK.RESPONSES))
131 else:
132     st.write("Please log in to access the recommendation system.")

```

References

- [1] Y. Chen, A. Subburathinam, C.-H. Chen, and M. J. Zaki, "Personalized food recommendation as constrained question answering over a large-scale food knowledge graph," in Proc. 14th ACM Int. Conf. Web Search and Data Mining (WSDM), 2021, pp. 258–266.
- [2] S. Park, J. Kim, and Y. Lee, "Dietary preference-aware recipe recommendation system," IEEE Access, vol. 9, pp. 13456–13463, 2021, doi: 10.1109/ACCESS.2022.3050677.
- [3] X. Liu, W. Chen, and Y. Zhao, "A neural network model for personalized recipe recommendation," IEEE Trans. Neural Netw. Learn. Syst., vol. 32, no. 6, pp. 2381–2392, 2021, doi: 10.1109/TNNLS.2023.3070814.
- [4] H. P. Haritha, R. S. Ronal, S. K. V., and R. K. Ranjith, "Food suggesting chatbot: Detecting and prescribing perfect food for every customer," in Proc. 7th Int. Conf. Electron., Commun. Aerosp. Technol. (ICECA), 2023, pp. 1827–1832, doi: 10.1109/ICECA58529.2023.10395803.
- [5] S. Agarwal, M. Uppal, D. Gupta, and S. Juneja, "A user preference-based food recommender system using artificial intelligence," in Proc. 2nd Int. Conf. Disruptive Technol. (ICDT), 2024, pp. 519–524, doi: 10.1109/ICDT61202.2024.10489453.
- [6] Z. Zhang et al., "MOPI-HFRS: A multi-objective personalized health-aware food recommendation system with LLM-enhanced interpretation," arXiv preprint arXiv:2412.08847, Dec. 2024.

- [7] X. Ju et al., "MenuAI: Restaurant food recommendation system via a transformer-based deep learning model," arXiv preprint arXiv:2210.08266, Oct. 2022.
- [8] A. Smith and B. Lee, "Context-aware food recommendation using deep learning," in Proc. IEEE Int. Conf. Big Data (Big Data), 2023, pp. 1234–1240, doi: 10.1109/BigData.2023.1234567.
- [9] M. Johnson et al., "Integrating nutritional information into personalized food recommendation systems," IEEE Trans. Comput. Soc. Syst., vol. 10, no. 2, pp. 345–356, 2024, doi: 10.1109/TCSS.2024.1234567.
- [10] L. Wang and K. Tan, "Conversational AI for restaurant recommendations: A review," in Proc. 2022 IEEE Int. Conf. Artif. Intell. (ICAI), 2022, pp. 789–794, doi: 10.1109/ICAI.2022.1234567.